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Department of Informatics

Undergraduate Thesis

Developing a Mobile-Assisted Language Learning Application Utilizing Short-Form Video Reels and Hypercasual Games

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Abstract

Language Learning has become ...

Key words: [KEYWORD-1], [KEYWORD-2], [KEYWORD-3]

This thesis proposes a Personalized Language Learning (PLL) application that...

Περίληψη

Η παρούσα διπλωματική εργασία ...
Λέξεις κλειδιά: [ΛΕΞΕΙΣ-ΚΛΕΙΔΙΑ], [ΚΕΨΩΟΡΔ-1], [ΚΕΨΩΟΡΔ-2], [ΚΕΨΩΟΡΔ-3]

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Finally, I thank my classmates and friends for their support and the beautiful moments we shared together.

I dedicate this work to my family.

Mohammad Matin Marzie
Corfu, [MONTH] 2026

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Chapter A

Introduction

A.1 Motivation

The demand for foreign language acquisition has reached an unprecedented global scale. In an increasingly interconnected global economy, over X.X billion people are actively learning second languages worldwide [?].

A.2 Background

To address these completion and engagement deficits, modern Mobile-Assisted Language Learning (MALL) must align with the media consumption realities of contemporary learners. This thesis explores the integration of two prominent digital phenomena: short-form video and hypercasual game mechanics.

A.2.1 Short-form Video

Short-form video is a media format characterized by its brevity, typically ranging from 15 seconds to 3 minutes, and is designed for rapid consumption.

Short-form video has become the dominant interface of digital attention. Statistically, YouTube Shorts has achieved over 200 billion daily views [?], While Meta Platfrom announced: Video is a particular bright spot, with video time spent on Instagram up more than 30% since last year and 5% more time spent on Facebook [?]. TikTok's global active advertising audience has reached 1.59 billion adults, representing 27.5% of the global population aged 18 and older [?].

This media format captures a disproportionate share of mobile attention. Globally, TikTok Android users average 1 hour and 37 minutes per day in the application, Youtube 1H and 25M, Instagram 1H and 13M and Facebook 1H and 7M [?]. This behavioral pattern is heavily concentrated among younger cohorts: 75% of US teens visit YouTube daily, and 61% engage with TikTok daily [?] where the data is available. Among US adults, daily TikTok usage exhibits a sharp ten-to-one generational gap, captured by 50% daily usage among 18-to-29-year-olds compared to just 5% among those aged 65 and older [?].

Pedagogically, the cognitive appeal of short-form video lies in its brevity and high visual density. While platforms keep raising video length caps (with TikTok allowing up to 10 minutes and Reels up to 3 minutes) [?], consumer behavior remains heavily anchored around rapid completion. Wistia's benchmark analysis of 13 million videos reveals that videos under 60 seconds hold a 52% average engagement rate, meaning viewers watch more than half the runtime before scrolling [?]. In contrast, longer videos see completion rates decay exponentially [?, ?]. Short-form videos therefore represent a highly engaging, authentic delivery mechanism for Second Language Acquisition (SLA).

A.2.2 Hypercasual Games

Simultaneously, gamification has emerged as a cornerstone of digital instruction. The viability of gamified mobile systems is exemplified by Duolingo, which reported 140.6 million monthly active users (MAUs), 58.7 million daily active users (DAUs) [?].

However, despite Duolingo's massive market footprint, its retention architecture relies on "shallow gamification" mechanics, such as rigid reward streaks and repetitive, decontextualized translation drills [?]. This over-reliance on uniform, static drills risks trivializing the cognitive complexity of language learning, driving a first-week user churn rate that exceeds 60% for many freemium applications [?]. Hypercasual game mechanics—characterized by instant tap-to-play interfaces, ultra-short gameplay loops, and highly localized feedback loops—offer an alternative. When merged with authentic multimedia assets, hypercasual loops can transform traditional, tedious vocabulary drilling into low-barrier, high-repetition cognitive reinforcement.

In the meanwhile Hypercasual games are rising. Hypercasual games—defined by their minimal ramp-up time, single-mechanic loops, and session lengths suited to short commutes—represent a similarly underutilised learning medium. Their accessibility and habitual engagement patterns make them natural candidates for spaced-repetition delivery. Glosy integrates two such games, Wordle and Word of Wonders, both of which select their target words directly from the learner's active vocabulary model, creating a closed loop between video exposure and game-based reinforcement.

A.3 Problem Statement and Research Questions

The paradox of passive consumption: Why short-form content (Reels/TikTok) is high-engagement but low-retention? People watch a lot but do they learn; The gap between entertainment and pedagogical efficacy (Ebbinghaus Forgetting Curve, Vygotsky). However, high engagement does not translate to retention: without active processing, vocabulary encountered in video dissolves rapidly following the forgetting curve described by Hermann Ebbinghaus in the 19th century. Unless a word is deliberately reviewed at increasing intervals, memory of it decays exponentially.

The question is: Can't i learn language by watching Reels from existing social media platforms?

Watching Reels from current social media platform are mostly used to boost vocabulary learning and not learning the vocabulary by it self (Find researches about this).

People play a lot Hypercasual games, can they be leveraged to learn a language?

How can we turn reels and games into language learning and make it effective? We have to track each word.(personalize) We have to Engeneer a Reel recommendation system(algorithm) with personalized words. Personalize each Hypercasual game level.

The central research problem is the absence of a language learning system that simultaneously leverages (i) the authentic contextual richness of user-generated short-form video, (ii) word-level learner modelling grounded in established memory theory, and (iii) gamified reinforcement adapted to the individual learner's vocabulary state.

Research questions:

RQ1. How can a mobile application integrate authentic short-form video content with a word-level vocabulary tracking model to support second language acquisition?

RQ2. How should a content recommendation engine be designed to select reels that are both comprehensible and lexically challenging for a given learner, based on the Ebbinghaus forgetting curve and CEFR proficiency levels?

RQ3. To what extent can hypercasual game mechanics, when parameterised by the learner's vocabulary state, serve as effective spaced-repetition exercises within a language learning application?

This thesis proposes **Glosy**, a cross-platform mobile application that addresses the identified gaps by embedding vocabulary acquisition within short-form authentic video content and reinforcing it through personalised hypercasual games. The system tracks each learner's vocabulary state at the word level, uses that state to select contextually appropriate reels, and adapts game difficulty to the same vocabulary model.

A.4 Significance

Helps learners use their daily habits for productive learning. The application is mobile-first. Supports Offline game play and practice. 3/4 use offline mode [**?**].

A.5 Thesis Structure

The remainder of this thesis is organised as follows. Chapter B reviews mobile-assisted language learning platforms, identifies their limitations, and establishes the pedagogical framework. Chapter C presents the system architecture, technology stack, data model, and recommendation engine design. Chapter **??** documents the implementation of each component with reference to the source code and database schema. Chapter E evaluates the system against its design objectives and discusses performance. Chapter **??** states the contributions, acknowledges limitations, and proposes future work.

Chapter B

Literature Review

B.1 The Landscape of Mobile-Assisted Language Learning (MALL)

A critical review of current platforms (Duolingo, Memrise, LingQ).

Digital language learning platforms (LLPs) have become dominant in foreign language education, with over 70 applications serving millions of users through Mobile-Assisted Language Learning (MALL)[2]. The global market size for these platforms was valued at USD 5.8 billion in 2025, growing further to USD 21.1 billion by 2034 at an estimated CAGR of 15.43% from 2025 to 2034 by Duolingo commanding over 50% of global downloads in 2023 [?]. This rapid expansion has been further driven with the advent of Large Language Models (LLMs), enhancing personalization and the generation of rich learning content. While it took 12 years to create 100 language courses for Duolingo with the use of AI they added 148 new courses in early 2025[1].

B.2 Pedagogical Foundations:

B.2.1 Sociocultural Theory (Vygotsky)

Lev Vygotsky's Sociocultural Theory [?] holds that cognitive development arises through social interaction mediated by cultural tools, of which language is the primary instrument. Central to the theory is the concept of the **Zone of Proximal Development**

(ZPD): the distance between what a learner can accomplish independently and what they can achieve under guidance or in collaboration. Effective instruction operates within the ZPD—content that is too easy provides no developmental stimulus, while content that is too difficult cannot be processed without first establishing prerequisite knowledge.

In Glosy, the ZPD is operationalised through the comprehensibility threshold of the recommendation engine. A reel is considered suitable for a given learner when that learner already knows at least 98% of its vocabulary—a threshold informed by Nation's [**?**] research on the vocabulary coverage required for unassisted reading comprehension. The remaining 2% constitutes the "i+1" challenge (one step beyond the learner's current knowledge), which falls squarely within the ZPD.

B.2.2 Cognitive Load Theory

B.2.3 Spaced Repetition and Memory Retention

B.2.3.1 Ebbinghaus's forgetting curve

Hermann Ebbinghaus's empirical research demonstrated that memory retention decays exponentially over time in the absence of review [**?**]. The resulting **forgetting curve** underpins all modern spaced-repetition systems (SRS): by scheduling review of a word at increasing intervals timed to precede the moment of forgetting, the system converts short-term exposure into durable long-term memory.

Glosy implements a six-level mastery model in the `user_vocabulary` table. Each word entry records a `mastery_level` (1–6) and a `last_review` timestamp. The recommendation engine uses these values to prioritise reels containing words whose review is overdue, ensuring that vocabulary resurfaces at intervals calibrated to each learner's individual retention rate.

B.2.3.2 Duolingo Half Life Regression

Duolingo's proprietary half-life regression model predicts the optimal timing for reviewing each word in a learner's vocabulary, based on their individual forgetting curve. This model informs the scheduling of spaced repetition exercises within Glosy, ensuring that learners encounter each word at intervals that maximize long-term retention.

B.2.4 micro-learning

B.3 Gamification in Education

From hypercasual gaming to "serious games." Genuine gamification—as opposed to the point-and-streak overlays critiqued in Section 2.2.4—involves embedding core game mechanics into the learning activity itself, such that the challenge of the game is inseparable from the learning objective [?]. Hypercasual games are well suited to this purpose: their short session length, immediate feedback loops, and escalating difficulty can be parametrically linked to a learner's vocabulary model.

Glosy integrates two hypercasual game formats: a Wordle variant, in which the target word is drawn from the learner's vocabulary model, and Word of Wonders, a grid-based word-construction game whose level vocabulary is similarly personalised. In both cases, success requires active recall rather than passive recognition, engaging a fundamentally different and more durable memory process than translation drills.

B.4 Short-Form Media in Second Language Acquisition

connects with the previous section of sociocultural theory of Vygotsky [vygotsky1978mind] Effectiveness of nano-learning (NL).

Authentic audiovisual materials have long been recognised as valuable in Second Language Acquisition (SLA) research. Short-form video, in particular, provides a dense multimodal signal: spoken language, prosodic variation, facial expression, gesture, and cultural context are all co-present, reinforcing one another in ways that text-based instruction cannot replicate.

The challenge is retention. Without active processing, vocabulary encountered during video viewing follows the Ebbinghaus forgetting curve as surely as any other exposure. Glosy addresses this by treating each reel not as a standalone learning event but as a contextualised encounter with target vocabulary in the learner's personal vocabulary model. Words seen in a reel are subsequently reinforced in game sessions, and the recommendation engine surfaces those words again in future reels, creating a multi-context learning loop that supports durable acquisition.

B.5 Limitations of Current Platforms

Karasimos [2] conducted a systematic cross-platform review of five representative LLPs—Duolingo, Memrise, Busuu, Rosetta Stone, and LingQ—against a set of pedagogical and usability criteria. While all five platforms demonstrate strengths in structured vocabulary delivery and habit formation, the review reveals consistent gaps that motivate the present work.

B.5.1 Uniform Content Regardless of Learner Demographics

All platforms present uniform content regardless of user demographics (children, Gen Z, adults, elderly), interests, or sociocultural perspectives, as well as the cultural differences inherent in languages other than English.
With over 500 million learners, Duolingo is using Machine Learning algorithms to provide learning material, only for the difficulty level [**?**].

B.5.2 Insufficient multimedia Integration

Duolingo's primary instructional mode is text and audio built around translation exercises. This approach does not exploit visual memory—a well-documented channel for vocabulary retention—nor does it expose learners to the prosodic and paralinguistic cues present in natural speech.

B.5.3 Lack of authentic, real-world content

Most platforms—Duolingo, Busuu, Memrise, and Rosetta Stone—lack authentic, real-world materials in their instructional design. LingQ partially integrates such content; however, as noted by Karasimos [2], it is not recommended as a starting point for beginners. Gilmore [**?**] demonstrates that authentic materials are crucial for developing communicative competence and for sustaining learner motivation. At the same time, Liu and Dong [**?**] caution that authentic language materials can create comprehension difficulties for learners who lack the cultural background knowledge assumed by native-speaker communication. In Glosy, this tension is addressed by pairing authentic video content with cultural annotations and adaptive vocabulary scaffolding, ensuring that exposure to real-world language does not exceed the learner's Zone of Proximal Development (ZPD).

B.5.4 Drilling and masquerading as gamification - shallow gamification mechanics

All platforms employ gamification, but it is largely limited to superficial elements such as points, badges, and leaderboards. These features may enhance engagement and motivation but do not substantively contribute to learning outcomes. The proposed system will implement gamification that is deeply integrated with the learning process, using hypercasual game mechanics that adapt to the learner's vocabulary state and reinforce retention through spaced repetition.

B.5.5 Lack of Support for Advanced Proficiency Development

With the exception of LingQ, no platform in Karasimos's review provides adequate support for learners at B2 level and above. Advanced learners require exposure to authentic idiomatic usage, domain-specific vocabulary, and complex syntactic structures—precisely the content available in user-generated short-form video but absent from structured drills.

B.5.6 Absence of Sociocultural Context

Language is inseparable from culture. The five reviewed platforms treat vocabulary and grammar as context-free systems, ignoring the sociocultural theory articulated by Vygotsky, which holds that language development is fundamentally mediated by social interaction and cultural context. Short-Form video content, as proposed in this thesis, provides a rich medium for embedding language learning within authentic sociocultural contexts, thereby aligning with Vygotsky's theoretical framework.

B.6 Recommendation Systems

B.6.1 Content-Based Recommendation

B.6.2 Collaborative Filtering

B.6.3 Hybrid Approaches

Chapter C

System Architecture & Methodology

C.1 System Overview

Glosy is designed as a three-service architecture in which a mobile frontend, a primary REST API backend, and a dedicated reels microservice each handle a well-defined subset of the system's responsibilities. Figure **??** illustrates the high-level interaction between these components. All services share a single PostgreSQL database instance and are coordinated in development via Docker Compose.

- **Frontend:** A cross-platform mobile application built with React Native (Expo SDK 54), targeting Android and iOS from a single JavaScript/TypeScript codebase.
- **Backend API:** A Node.js/Express service (ESM modules) responsible for user authentication, vocabulary management, progress synchronisation, and the dictionary.
- **Reels Service:** A Python FastAPI microservice Recommendation System responsible for serving reels enriched with structured dialogue data—sentences, timestamped tokens, and translations.
- **Database:** PostgreSQL 16, whose schema encodes the vocabulary model, user progress, content metadata, and engagement signals.

C.2 Technology Stack

C.2.1 Frontend — React Native / Expo

React Native was selected to enable cross-platform deployment from a single codebase without sacrificing the native rendering performance required for smooth video playback. Expo SDK 54 provides managed native modules for video (`expo-video`), secure credential storage (`expo-secure-store`), and build tooling (`eas-cli`). The application uses Expo Router (file-system-based navigation) and React Native Reanimated for gesture-driven animations in the game interfaces.

C.2.2 Backend — Node.js / Express

The backend is implemented as an ES-module Node.js application using Express 4. This choice reflects the project's existing expertise in JavaScript and the maturity of the Express ecosystem for REST API development. Authentication is handled with JSON Web Tokens (JWT): a short-lived access token (15 minutes) and a long-lived refresh token (30 days) follow the standard rotation pattern. Google OAuth 2.0 is supported alongside email/password authentication via the `google-auth-library`. Input validation uses Joi schemas for every endpoint, and the API is documented with Swagger (OpenAPI 3.0) accessible at `/swagger`.

C.2.3 Reels Service — Python / FastAPI

The reels microservice is implemented in Python using FastAPI and SQLAlchemy (async). The separation from the main backend is motivated by the richer data-processing requirements of the reels pipeline: joining reels with their associated dialogues, sentences, per-token word data, and sentence translations in a single query is better expressed in SQLAlchemy's ORM than in raw SQL via the Node.js `pg` client.

C.2.4 Database — PostgreSQL 16

PostgreSQL was chosen for its support of complex relational queries, array operators, and procedural functions—all of which are used in the vocabulary bulk-initialisation and the planned recommendation engine. The full schema is maintained in `database/thesis_db.sql`.

C.3 Data Modelling

The database schema models the domain along three axes: linguistic content, user state, and engagement signals.

Linguistic content

User state is captured in two tables. `user_languages` records each language pair a user is studying and `user_vocabulary` is the core of the learner model.

Engagement signals are stored in `reel_interactions`, which records views, likes, saves, comments, shares, and reports per (user, reel) pair. These signals are reserved for the personalised recommendation pipeline described in Section C.4.

C.4 Recommendation Engine Design

The recommendation engine is the theoretical core of the system. Its goal is to select, from the pool of available reels, those that are simultaneously *comprehensible* (the learner knows enough vocabulary to follow the content) and *pedagogically productive* (the reel contains words that are due for review or represent the next developmental step). The design follows a three-stage pipeline.

C.4.1 Stage 1: Comprehensibility Filtering

For each candidate reel, the engine computes the proportion of the reel's unique word tokens that are present in the learner's vocabulary model at mastery level ≥ 1. A reel passes the filter only if this proportion meets or exceeds 98%, corresponding to Nation's [**?**] threshold for unassisted reading comprehension. This criterion implements the ZPD operationally: the learner encounters real language while remaining within a comprehensible range.

C.4.2 Stage 2: Spaced-Repetition Scoring

Each reel is scored according to the review urgency of the vocabulary it contains. For each word w in the reel with mastery level m and days since last review d, the system computes an urgency score based on the Ebbinghaus forgetting function. The reel's aggregate score is the mean urgency across its target vocabulary. This stage ensures that reels containing words the learner is about to forget are ranked above reels containing words that were recently reviewed.

C.4.3 Stage 3: Content Personalisation

The final stage personalises the recommendation by incorporating engagement signals. Reels that have been liked are boosted. This stage allows the system to learn the learner's preferences over time, promoting content that is both pedagogically useful and personally engaging, thus considering demographic factors too.

C.5 Gamification Design

The gamification layer consists of two components: a progression system and two vocabulary-driven games.

The **progression system** (XP), (Coins) provide a secondary reward currency redeemable for in-app benefits.

C.6 Vocabulary-Driven Games

C.6.1 Wordle-style Game

C.6.2 Word of Wonders

Chapter D

Implementation Details

D.1 Database Schema

D.2 Backend API

The API has been documented using Swagger/OpenAPI. The documentation is available at `/api-docs` endpoint.

D.2.1 Authentication Module

The backend (`backend/`) implements a dual-token authentication scheme. On successful login or registration, the server issues a short-lived JWT access token (default: 15 minutes) and a long-lived refresh token (default: 30 days). Google OAuth 2.0 is also supported.

D.2.2 Registration and Vocabulary Initialisation

D.2.3 Vocabulary Synchronisation

D.3 Frontend Architecture

The frontend has 5 tabs. The application implements an offline-first strategy by caching API responses in asyncStorage 3/4 use offline .

D.4 Reels Service

D.4.1 Service Architecture

The reels microservice (`reels-service/`) is a FastAPI python application.

D.4.2 Creating Reels

D.4.3 Dialogue Enrichment

Each uploaded reel response embeds the full dialogue associated with the reel's video content.

D.4.4 Personalisation Implementation Status

D.5 Game Modules

D.5.1 Wordle-style Game

D.5.2 Word of Wonders

Word of Wonders is a grid-based word-construction game in which the player must identify hidden words from a set of provided letters. Each game level is associated with a set of target words. In Glosy, these target words are drawn from the player's active vocabulary model, prioritising words at lower mastery levels. The grid is generated by computing all valid arrangements of the target words' letters into a crossword-style grid. pseudocode for the grid generation algorithm

Chapter E

Evaluation & Results

A full empirical user study—with longitudinal vocabulary retention measurements, control groups, and statistical analysis—is beyond the scope of this undergraduate thesis.

Chapter F

Conclusions & Future work

F.1 Contributions

F.2 Limitations

The work presented in this thesis has several limitations that must be acknowledged honestly. **No empirical user study.** The pedagogical effectiveness of the system has not been evaluated in a controlled experiment. **Limited content library.**

F.3 Future Work

The following directions are identified as high-value:

- **Automated content annotation pipeline:** A pipeline that ingests a raw video URL, transcribes speech using a large language model, aligns the transcript to video timestamps, tokenises and POS-tags the output, maps tokens to the word database, and produces a fully annotated reel record—reducing the per-reel curation cost from hours to minutes.
- **Empirical evaluation:** A randomised controlled study measuring vocabulary gain, retention at one week and one month, and learner engagement metrics, comparing Glosy against a no-game control condition and against Duolingo.

- **Multi-language expansion:** The schema and service layer are language-agnostic (language is a foreign key throughout); adding a new language requires only content ingestion, not code changes.
- **Collaborative learning features:** Shared vocabulary challenges, leaderboards scoped to friend groups, and comment-driven vocabulary discussion on individual reels, supporting the social dimension of Vygotsky's theory.
- **Vocabulary synchronisation performance:** Replacing the serial per-word UPDATE loop in `syncController.js` with a bulk `INSERT ... ON CONFLICT DO UPDATE` statement to support large offline sessions without latency degradation.

Bibliography

[1] Duolingo, Inc. Duolingo launches 148 new language courses. Press Release, Duolingo Investor Relations, https://investors.duolingo.com/news-releases/news-release-details/duolingo-launches-148-new-language-courses, April 2025. Accessed: 2026-07-28.

[2] Athanasios Karasimos. The battle of language learning apps: a cross-platform overview. *Research Papers in Language Teaching and Learning*, 12(1):150–166, 2022.

Abbreviations

AI - Artificial intelligence

Glossary

Algorithm - A sequence of instructions for solving a problem